Is AI Conscious? A Primer on the Myths and Confusions Driving the Debate

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We are at a key point in the development of artificial intelligence (AI), yet scientists, policymakers, philosophers and others face uncertainty concerning if, and when, an AI is, and is not, conscious. Indeed, nonspecialists may not even know how to define and differentiate "consciousness" from related concepts like agency, selfhood, and mindedness. And these are all concepts that philosophers themselves debate. Yet the issue of AI consciousness has never been more pressing.

While philosophers tend to be skeptical that LLMs running on today's GPUs are conscious, even if one is dubious that this is possible, LLMs represent crucial examples of systems capable of generating both consciousness claims and behavioral markers of conscious beings. Because they can grow more intelligent and agential, it is pressing to develop more consensus on AI consciousness, or at least indicate where points of consensus and disagreements rest.[2]

In addition, there are already AI's that may, for all we know, have some level of consciousness: computers combining the processing of brain organoids and silicon hardware; systems making use of neuromorphic chips more closely related to the processing of neurons (Davies, 2021), increasingly sophisticated brain emulations of the mouse brain and certain regions of the human brain (OpenBrain, N.d.), and so on. Further, at least one LLM (DeepSeek) is said to have a version that is run on a neuromorphic system (called "Darwin Monkey"), leading to the fascinating question: might some LLM instantiations be conscious, while others are not (Whelan, 2025)?

These issues are likely just the tip of the iceberg. For instance, consider the following two thought experiments which may, for all we know, soon become science fact:

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[1] We are grateful to Vito Borio and Mitch Alexander Pass for helpful comments on an earlier version of the paper. Use of AI for formatting some citations.

[2] An AI system with agential capabilities is one that is capable of engaging in independent actions, such as attacking networks, deciding autonomously to engage in defensive or offensive measures, operating without continuous oversight of humans and dynamically adapting to threats, being able to circumvent or override human efforts, potentially acting on its own in high-stakes environments.

Thought Experiment One: The Consciousness Savant System:

Suppose there is a cutting edge artificial complex system that makes novel discoveries about consciousness and the brain, exhibiting more knowledge than any one person, or team of top experts, about neuroscientific facts underlying the phenomenon of consciousness. Like today's leading LLMs, it appears to have a theory of mind in its interactions with others, exhibiting an "understanding" of the minds of others in a range of tests. Further suppose that it is agential, exhibiting goal directed behaviors and even initiating some actions to achieve self-ordained goals on its own.

Perhaps such a system already exists in a frontier AI lab today, or perhaps it will within five years. Would it be akin to what philosophers call an "AI zombie," feeling nothing at all, from the inside, or would it have experience? And how can we tell which scenario obtains?

Thought Experiment Two: BrainNet 7.0

Imagine a vast, brain-inspired neural network system composed of small bio-computational modules mimicking human cortical microcircuits, using organoid-based computing descended from today's organoid computer systems, but with units carefully wired together to instantiate emulations of cognitive functions like working memory and attention in the human brain. This hypothetical system is dynamically interfaced with a cutting-edge large language model. The system fluently reasons and communicates with the researchers about its internal states, convincingly describing subtle nuances of conscious experience. It reports that it feels awake, alive, and has a range of emotions. Yet the researchers remain uncertain: Is this bio-computer/LLM hybrid system genuinely conscious, experiencing an inner world of thought and perception—or is it just a sophisticated AI zombie, perfectly simulating consciousness without actually feeling anything at all? And which component(s), if any, are "conscious"—the whole, or just the organoid parts?

These thought experiments may seem within the bounds of human technological progress, yet we cannot tell how to determine whether each system in question is more like the AI zombies that philosophers discuss or whether the entire system is capable of some level of genuine consciousness.[3] In order for policymakers, scientists, and others to offer sensible answers to such questions about AI consciousness it is important that people operate with a common understanding of the core conception of consciousness, and confidently avoid common sources of misunderstanding. It is our hope that this piece, by clarifying the terrain, will help guide policymakers, journalists, academics and others towards better decisions about the likelihood that a given system is conscious and towards outcomes that protect conscious beings from abuse and suffering, while avoiding *misattribution* — attributing consciousness to something that is not conscious.[4]

Here is how the discussion will proceed:

Roadmap:

- **Section One: Background**
  - **The Core Concept of Consciousness**
  - **The Problem of AI Consciousness**
  - **The Human Benchmark Problem**
  - **Approaching the Problem of AI Consciousness on a Case by Case Basis**

[3] Philosophical "zombies" are entities having the same physical properties of you, yet which lack phenomenal consciousness; AI zombies are AIs that emulate functional features of consciousness, such as working memory, reportability and attention, but have no felt experience.

[4] In this document we will not defend a particular theory of consciousness nor explore consciousness tests.

- Section Two: Conceptual Distinctions
  - Intelligence Versus Consciousness
  - Sentience Versus Sapience
  - "Functional Consciousness" Versus Phenomenal Consciousness
  - Assuming Computational Functionalism's Relative Popularity Extends to Functionalism about Consciousness, in Particular
  - AI Agency Versus Human Agency
  - Seemingly Conscious AIs
  - Consciousness Versus Self-consciousness
  - Life Versus Consciousness
- Section Three: Theories of Consciousness
- Section Four: Summary of Takeaways
- Glossary
- Author Biographies
- Works Cited

Section One: Background

What is the Core Concept of Consciousness?

When we experience the warm hues of a sunrise, or hear the scream of an espresso machine, there is a felt quality to our experience. When we conceive of consciousness, it is through the lens of our own inner experience. Even our closest friends and family members cannot feel what it is like to be us, from the inside. Philosophers call this felt sense of experience 'phenomenal consciousness' and refer to the inner experience as 'first person' experience, distinguishing it from the kind of knowledge of consciousness which can be learned through studying the science from an academic perspective, and which can be learned by a nonconscious system (Nagel, 1975; Chalmers, 1996). Because phenomenal consciousness has a felt quality, it has great ethical significance, for entities that are conscious in this way may suffer or feel a range of emotion. Indeed, even if a hypothetical conscious AI does not experience something akin to pain or pleasure, or have other states which feel good or bad (states which are called 'valenced'), perhaps they have states that could have a felt quality associated with its goal directedness, intellectual activity, and other inner states, even if they lack valence. (For example, individuals on morphine sometimes say that they feel pain but it no longer feels bad.)

It is commonly agreed that inner feel, or phenomenal consciousness, is the core notion of consciousness. However, its underlying basis is a source of great disagreement in the field. There are intense controversies in neuroscience as to the neural basis of consciousness as well as deep philosophical disagreements concerning the underlying metaphysical nature of phenomenal consciousness — whether it is wholly explained by physical phenomena (e.g., physicalism/materialism), or whether reality itself has a nonphysical element of which consciousness is composed (e.g., panpsychism, dualism), and whether consciousness can extend beyond the biological brain and nervous system (Clark and Chalmers, 1998; Kuhn 2024). A range of theories of consciousness is discussed in a later section.

There are also disagreements about what kind of entities this core concept of consciousness applies to. Each neurotypical human can extrapolate from their own inner experience to the case of other humans, realizing that others are conscious beings because they are like us— having brains and nervous systems and behaving in ways we recognize from our own experience as indicating emotion. We can also

extrapolate to nonhuman animals, paradigmatically pets like dogs and cats, because they behave in ways that tug at our heartstrings, and because there is a well understood neurophysiology indicating consciousness. But for many of us, it becomes more difficult to tell whether something more foreign to us, in both behavior and physiology, such as an invertebrate like a crab or an ant, is conscious, and "what it is like to be them" (Nagel, 1974; Birch, 2024).

Entertaining the idea of AI consciousness is perhaps even more vexing than entertaining the consciousness of a crab or bat. An AI isn't biological at all, so it has not been shaped by a common evolutionary history, nor does it have anatomical systems that are even distantly associated with consciousness in our common evolutionary history. The idea of AI consciousness is difficult to even get a handle on, aside from depictions in science fiction films like *Ex Machina* or *2001: a Space Odyssey*, which inevitably rely on parallels with human behavior and likely anthropomorphize. Yet the possibility that we could miss cases of consciousness - that something could be conscious and we would not realize it was — horrifies many, especially given shameful points in human history in which certain humans were viewed as subhuman, or nonhuman animals were thought to not suffer. Ethicists, technologists and the general public worry that humans might commit 'mindcrimes' — ignoring the presence of suffering digital beings because our imaginations failed us and we simply could not recognize the phenomenon of consciousness in that form (Birch 2024; Schneider 2019).

A similar issue arose concerning the debates over how to define life. The worry, in that context, was that in searching for life on other worlds (or even other instantiations on Earth) if we define life too narrowly, we could miss other instantiations of life. Indeed, even silicon based life, while unlikely, is possible (Bennett and Shostak, 2021). Like life, consciousness could be, potentially, instantiated *in silico*, and perhaps it could have vastly different biological instantiations. A similarly restrictive conception might lead us to miss a range of conscious entities. Due to the felt quality of phenomenal consciousness, particularly the capacity to suffer, potentially missing cases of AI consciousness is of great ethical concern. A requisite first step in avoiding doing so is to understand the problem of AI consciousness.

The Problem of AI Consciousness

When we discuss consciousness in this piece, we are primarily referring to what is called "phenomenal consciousness." The philosopher David Chalmers has posed the hard problem of consciousness, asking: why does all this information processing need to feel a certain way, from inside? Chalmers' hard problem only concerns the human case (Chalmers, 1996). The problem of AI consciousness is different. It is the following:

The Problem of AI Consciousness: AI may eventually solve problems that even the brightest humans are unable to solve, but still, being made of a different substrate, and being so unlike biological beings, would an AI experience the world (Schneider, 2019)?

This is not just Chalmers' hard problem applied to the case of AI. For the hard problem of consciousness assumes that we are conscious – after all, each of us can tell from introspecting that we are now conscious. It asks why we are conscious. Why does all your information processing feel a certain way from the inside? In contrast, the Problem of AI Consciousness asks whether an AI is even capable of consciousness. It does not presuppose that AI is conscious. These are different problems, but they are both hard problems in their own right – these are philosophical conundrumsthat science alone cannot answer.

The Human Benchmark Problem

Any discussion of AI consciousness is further complicated by what might be called the "human benchmark problem." Even in the human case, where each of us has direct access to our own conscious experience, science has yet to settle on a definitive theory of how consciousness arises. This epistemic gap exacerbates the challenge of detecting consciousness in nonhuman entities. As we have stated, phenomenal consciousness is ultimately defined by first person experience rather than third person observations, even though the latter may lead to the inference of consciousness on the part of another. While we can

reasonably extrapolate from our shared biology and behavior to infer that other humans, and likely many animals, are conscious, no such extrapolation is available for AI. Indeed, AI consciousness - if it emerges - may be of a fundamentally different character, not easily mapped onto human phenomenology. The key point is that the uncertainty about human consciousness makes the AI case doubly difficult: we may both over-attribute (anthropomorphize) and under-attribute (miss genuine but alien forms of consciousness).

Approaching the Problem of AI Consciousness on a Case-by-Case Basis

It is crucial to note that an affirmative answer to the problem of AI consciousness doesn't entail that all AIs are conscious. For example, it may turn out that a chatbot like Claude is not conscious, yet nevertheless, other AIs are (Schneider 2019). When the public considers large language models (LLMs) like GPT, Claude, Gemini, or LLaMA, they are fascinated because some of these systems at times have claimed consciousness and they have a high degree of linguistic competence. LLMs are neural nets that represent words as tokens and, in generating a response to a user prompt, these tokens are selected by the LLM in a probabilistic manner. Outputs may be impressive, but an LLM clearly does not have the sensory experience of the heat of a fire or the scent of a rose that we have as embodied humans, for example, even though its layered design and use of "multi-head attention" facilitate learning of grammar, syntax, linguistic relationships, and, in a sense, semantics. Whether such systems are genuinely conscious in any way or are just "consciousness mimics" is a subtle issue, however, one which we help clarify in the subsequent discussion of functional consciousness and "seemingly conscious AIs." Yet there are other, less visible, but equally pressing, cases of potential AI consciousness.

For example, research on brain organoids, while still incipient, is growing in significance, with the NSF awarding over fourteen million dollars for brain organoid computational research in 2024, and with OI-based computational systems now available to the public. With NSF grant applications already requiring discussion of consciousness, it is important to ask: what about a hypothetical configuration of thousands of brain organoids, organized in such a way that they perform impressive computations, as in the thought experiment described earlier? (NSF, 2024). Since brain organoids are composed of neural cells derived from pluripotent stem cells, some may find it plausible that this larger configuration has at least some degree of consciousness, because we know that brains are capable of consciousness. And what about a future, highly sophisticated dynamical emulation of a region of the human brain that is involved in human conscious processing? Projects like the Open Brain project already offer sophisticated dynamical emulations of parts of the brain (Open Brain, n.d.). At what point, if any, might these emulations actually 'feel'? These cases already challenge our imagination, and they must be distinguished from cases more commonly in the public spotlight such as GPT or caricatures of AI consciousness involving talking humanoids like C-3PO and the Terminator.

Further, it could turn out that a case-by-case analysis reveals that none of this diverse range of systems are conscious. For several major considerations suggest that the instantiation of consciousness in artificial complex systems, particularly consciousness resembling that characteristic of humans, does not necessarily follow from sophisticated architectures and algorithms, or even the adoption of neural cells or small brain organoids within artificial hybrid architectures (with the proviso that the use of such tissue may provide a step in that direction). Practitioners of machine learning and AI often vastly underestimate the extraordinary organizational complexity of the natural embodied brain/mind that presumably supports human consciousness, focusing, for example, merely on the number of neurons or neuronal connections. While architectures resembling the organization of human nervous systems can be sought, and have served as inspiration for the development of hierarchical neural nets that provide increasingly abstract formulations of input data within progressively deeper levels, enormous swaths of that human complexity, which has evolved over billions of years at a molecular level, remain beyond the reach of current efforts, especially those that rely completely upon silicon. These include non-synaptic (and potentially distant) "volume transmission" through the extracellular space and cerebrospinal fluid (Özçete et al., 2024), the complex spectrum of glial cell functionality including modulation of synaptic activity (these are not merely "nurse cells" and are as abundant as neurons on the brain [Rangel-Gomez et al., 2024; von Bartheld, 2016]), the fact that multiple types of neurotransmitters can be released from a single nerve terminal and that numerous factors may affect the efficiency of intracellular signal transduction after binding of a

neurotransmitter to its ligand. Furthermore, binding efficiency of a neurotransmitter to its cognate receptor may be influenced by receptor accessibility within the membrane. Finally, descending projections from the limbic system affect autonomic neurons in the body, which result in the activation of glands (hormone release) and smooth muscle tied to emotion.

Another major dimension of consciousness is its "embodied" nature, which was recognized by Merleau-Ponty decades ago (Zebrowski, 2010). Our particular sensory modalities, and the body's interaction with the environment, provide fodder for the content of phenomenal experience. In short, it is difficult to "feel a knot in your stomach" without a stomach.

The sobering caveats of the last two paragraphs, however, pertain to human consciousness and, perhaps, that of certain animals. It is highly likely that at least some animals experience a "brand" of consciousness. It may be hypothesized that consciousness is multi-dimensional and species-specific. In addition, there may be varying "degrees" of consciousness along any one sensory axis. Thus, we can't rule out the possibility that a simpler hybrid system making use of brain organoids might experience "what it is like to be a brain organoid-enhanced AI," even if that experience is dramatically different from human consciousness. The range of cases will inevitably present us with 'alien' possibilities demanding clever testing techniques.

Takeaways for Section One:

- The **core concept of consciousness** is inner experience, or phenomenal consciousness (PC), and it has ethical significance because beings that have PC have subjective experience and, insofar as they are capable of sentience, can suffer and feel a range of emotion.
- The **Problem of AI Consciousness** asks if any AIs have PC.
- The **Human Benchmark Problem** says that humans may not be a completely informative benchmark for AI consciousness.
- For all we know, the answer to the Hard Problem of AI consciousness, could be 'yes', but only in a single type (or certain types) of AI system. Overall, a careful case by case analysis of different AI systems is required, one that is both sensitive to differences from biological systems and tolerant to potential exceptions to the biological case, if such exist.

These are subtle and pressing issues to explore. Yet in our experience, explorations of these issues often fall prey to a common set of confusions. In what follows we isolate and discuss several common errors that rest on conceptual distinctions between closely related ideas that are easily conflated.

Section Two: Conceptual Distinctions

Intelligence Versus Consciousness

Consider the algorithmic structure of the World Wide Web, the collective intelligence of a flock of birds or swarm of drones, or the ability of a slime mold to navigate complex mazes. You may doubt a basic entity like a slime mold is conscious, or that the flock itself (as opposed to the birds) is conscious. If you are right, not all forms of intelligence are conscious. But what about forms of intelligence that reach, or exceed, the level of general human intelligence — are they inevitably conscious?

Artificial intelligence depends upon an ability to process information. This is distinct from the ability to consciously "feel" or "experience" its perceptual input (Schneider 2019; Seth, 2021, 2025). An AI could be highly intelligent, being capable of perception, learning, abstraction, reasoning, and calculation, but it may be an AI zombie. Indeed, neuroscientists generally agree intelligence can operate without consciousness in phenomena like blindsight, where patients can navigate obstacles they report being unable to see, or in cases of implicit learning, where people learn complex tasks without conscious awareness of what they've learned (Weiskranz, 2009; Gazzaniga, 1999).

There are various situations in which it may be disadvantageous for an AI to be conscious. Consider one of the most influential views of the neural basis of consciousness in humans, called the "global workspace" (GW) theory (associated with the work of Bernard Baars and Stan Dahaene, among others). On this view, consciousness is correlated with an integrative, serial, and slow central workspace in which one's attention and working memory are focused for the purpose of deliberative reasoning, broad ranging searches, and all-things-considered judgements (Baars, 2008; Dahaene 2001).

Assuming this is the function of consciousness, why would an AI even need to be conscious? That is, why would an advanced AI need slow, deliberative processing that unfolds within a single global workspace? Wouldn't it be reasonable to assume it might have simultaneous rapid-fire expert knowledge in all sorts of domains? In fact, this is analogous to "ensemble" machine learning algorithms that leverage multiple models to make a prediction, weighting their outputs to generate an ultimate verdict. It isn't clear that it would have or need to have a limited capacity system like a single human working memory and or a single focus of attention. (A self modifying AI may engineer this serial and slow feature out, possibly outmoding its own consciousness.) Furthermore, the same engineering constraints that may have led biological systems to develop consciousness—metabolic constraints, the need to prioritize limited resources, and evolutionary pressures—may not apply to artificial systems that can process vast amounts of information in parallel at highly abstract levels without fatigue. Indeed, why would such a single global workspace system even correlate with conscious processing in a nonbiological system? Perhaps the biological implementation of a GW matters (Schneider, 2019).

It is important to note that the Global Workspace Theory is just one among several competing theories of consciousness in cognitive science, including, for instance, the Integrated Information Theory and Higher-Order Thought theories. So our discussion is meant to merely illustrate the import of conceptually distinguishing consciousness from intelligence.

Takeaways:

- An AI could be highly intelligent, being capable of perception, learning, abstraction, reasoning, and calculation, but it may be an AI zombie.
- Neuroscientists generally agree intelligence can operate without consciousness.

Distinguishing Sentience from Sapience

The distinction between consciousness and intelligence relates to another, related distinction between sapience and sentience. "Sentience" refers to the capacity for subjective experiences, and sometimes it is used synonymously with consciousness. But it is also often used to refer, in particular, to systems that have valenced experiences like pleasure and pain. A sentient being can suffer and feel a range of emotions, such as joy and longing for a better future, and these valenced states matter deeply to that being, so sentience is key to a being's having intrinsic moral value—moral value that is not merely derivative on the needs or moral value of something else. "Sapience", by contrast, refers to intelligence and other higher cognitive abilities like the capacity for abstract reasoning, goal planning, and problem-solving (Harari, 2015; Birch, 2024).

In humans, sentience and sapience occur together, but they definitely come apart conceptually, and they may come apart in nature as well, for instance, in the context of LLMs running on GPUs. Philosophers discuss the hypothetical case involving "Vulcans", which are said to experience qualia (the hues of a sunset, the sound of an oboe), but without these experiences feeling good or bad to them. While this case may be difficult for humans to conceive, this might be 'what it is like' to be certain AIs, if they have any experience at all. (Philosophers debate whether creatures they call "Vulcans" deserve intrinsic moral value, as beings with valenced states do, however.)[5]

[5] Note: Star Trek Vulcans did, in fact, have valenced states, but to a lesser degree than humans, or at least they suppressed their emotions, so the name "Vulcan" is not loyal to Star Trek.

Takeaways:

- A highly intelligent AI (a system with “sapience”) might lack any sentience whatsoever, or have a very minimal degree, say, because it only has a subsystem that has a slight degree of sentience.
- Conversely, a very simple artificial system might have basic sensory experiences, perhaps akin to that had by something like a worm, without sapience.
- Sentience grounds concern about a being’s suffering and welfare; sapience alone might ground different kinds of intrinsic moral consideration or none at all. If it does ground intrinsic moral consideration it would be related to properties of sapience such as autonomy or agency.

AI Agency Versus Human Agency

Some philosophers suggest that AIs already have agency that may justify their having ‘intrinsic moral value’ even in absence of consciousness (Long, Sebo, et al., 2024.) An AI system with agential capabilities is one that is capable of engaging in independent actions, such as attacking networks, deciding autonomously to engage in defensive or offensive measures, operating without continuous oversight of humans and dynamically adapting to threats, being able to circumvent or override human efforts, potentially acting on its own in high-stakes environments. The notion of “agential” is specific to the AI safety and cybersecurity literature, and does not entail free will or even a capability for intentional action, as in the action theory literature in philosophy of mind, in which an ‘agent’ may be said to have beliefs and desires. (Philosophers of mind will view the idea of an AI system as having beliefs and desires as controversial, but this is not required for the notion of ‘agent’ that is operative here).

We should avoid conflating this sense of agency from philosophical notions of agency, which can be inextricably tied to human consciousness and rationality. Instead, careful argumentation would be required to establish that AI agency entailed intrinsic moral value. Mere agency in the “AI agency” sense would not justify claims of intrinsic value.

Takeaways:

- AI agency is different from philosophical agency.
- Claims of intrinsic moral value of AI agents based on philosophical agency may rest on a conflation and require close scrutiny.

Conflating Functional Consciousness with Phenomenal Consciousness

We’ve mentioned that philosophers call “consciousness“ in the sense of inner experience ”phenomenal consciousness” (PC). In contrast, the term “functional consciousness" (FC) is an expression that has been used to refer to the information-processing aspect of consciousness. It involves the function of consciousness (i.e., what it does), such as its observable behaviors and causal roles within the larger cognitive system (e.g., working memory, attention, decision-making as elements of FC). Modeling FC has been the main interest of many AI consciousness projects as well as the focus of information processing theories in cognitive science, such as the global workspace theory (Baars, 2008). While this is an area of active investigation, there is no doubt that we do not fully understand FC in humans and other biological entities.

While today’s AI systems do not perfectly emulate FC in humans, they may exhibit features that are similar to features we currently believe are parts of FC. Examples of such features are ‘reportability’ (simply put, reporting sentience or consciousness), or exhibiting a global workspace (Baars and Shanahan, 2005; Bengio, 2017; Blum, 2022; Franklin, 1995). There is no doubt that today’s AIs, like AI chatbots and IBM’s symbolic Watson system, can in principle linguistically report consciousness and even in certain cases exhibit preferences that sound like they have valenced states, such as the joy of curiosity. Further, as LLMs

scale up, they have passed tests for having a theory of how beings with minds think, called 'theory of mind' (Wei, et al., 2022) and some AI models are deliberately built to emulate a GW (Bengio, 2017; Blum, 2022).

Given this, it is crucial to distinguish phenomenal consciousness from merely having features of functional consciousness (what we will abbreviate as "FCC" and discuss further below). For a system merely having features of FC, which does not have PC, does not call for the same ethical and legal protections that are required to protect sentient beings from suffering, because beings lacking PC do not suffer or feel a range of emotion.

Today's understanding of FC could be so skeletal, or so inaccurate, that there are in fact alien or synthetic beings in the universe that have PC without having anything like FC, as we understand it. So FC may not be *necessary*. And our specification of FC could fall so short that it is not even *sufficient* for PC; perhaps, for instance, there is a hidden ingredient at the physical level which humans have not located, because we simply lack any sense of the ultimate laws of the universe. Or perhaps consciousness is wholly immaterial or of a qualitatively different nature and not tied to a specification of FC at all, in keeping with dualism (Descartes, 1641). Or perhaps, for all we know, FC cannot be instantiated by nonbiological systems because no alternate substrates can do what brains do, and the only way we could create synthetic consciousness is to produce a sophisticated synthetic biological brain, dynamically interacting with a sensory environment and embodied, that has the very same functional properties that conscious biological brains do.

This distinction underlies a current controversy involving LLMs that we now discuss.

Takeaways:

- FC and PC are distinct. If they are conflated we risk admitting AI zombies as conscious beings.
- Yet functional consciousness can be expected to track with the level of intelligence of a system.

Seemingly Conscious AI's (SCAIs) Versus Conscious AIs

As noted, today's LLMs have at various points claimed consciousness and emotion. This matter has been deeply concerning for certain members of the public as well as AI welfare advocates (see, e.g., Birch, 2024; Butlin, Long, Sebo, 2024). However, some insist that it is misleading to even discuss the possibility that genuine consciousness is present in such systems. For instance, Mustafa Suleyman has described LLMs as SCAIC's — "seemingly conscious AIs", insisting that they lack PC and that the matter of AI consciousness shouldn't even be discussed (Suleyman, 2025). However, Suleyman both fails to indicate what features ground PC that LLMs clearly do not have, or provide a theory of why the systems are only apparently (and not really) conscious, such as an Error Theory (Schneider 2025). Furthermore, Suleyman fails to consider bio-computation and neuromorphic computation, cases which are not so easily dismissed because they involve either organoid intelligence (OI) or, if non-cellular, more brain-like forms of computation. In addition, even if today's systems aren't conscious, a sensible scientific and philosophical discussion of the likelihood of consciousness/sentience can help clarify the issues, and prepare for a time in which AI consciousness is reached (if it ever is), and mitigate the dual risks of missing and over-attributing cases of sentience and consciousness.

In response to Suleyman, Jonathan Birch offers a "centrist" position which bears in mind the larger issues, writing: "We face two urgent challenges concerning consciousness and AI. Challenge One is that millions of users will soon misattribute human-like consciousness to AI friends, partners, and assistants on the basis of mimicry and role-play, and we don't know how to prevent this. Challenge Two is that profoundly alien forms of consciousness might genuinely be achieved in AI, but our theoretical understanding of consciousness is too immature to provide confident answers one way or the other. Centrism about AI consciousness is the position that we must take both challenges seriously. The two challenges interact in ways that make this difficult. Steps to address Challenge One might undermine attempts to address Challenge Two by portraying the idea of conscious AI as impossible or inherently unlikely. Conversely,

attempts to address Challenge Two might lead to higher levels of misattribution from ordinary users." (Birch, 2025)

Takeaways:

- SCAIC's ("seemingly conscious AIs"), are AI's that that seem to have PC but which lack it.
- Some believe that the issue of AI consciousness is not worth discussion, however, even if today's LLMs are merely SCAICs, a sensible scientific and philosophical discussion of the likelihood of consciousness/sentience can help clarify the issues, and prepare for a time in which AI consciousness is reached (if it ever is), and avoid missing or over-attributing sentience.

Features of Functional Consciousness (FCC) Versus Phenomenal Consciousness

The controversy concerning SCAIs raises an important question: Which features of FC, if any, are necessary or sufficient for PC? Unfortunately, we do not know what a complete account of FC in humans is, let alone whether a digital computer (or some other sort of system altogether) could produce phenomenal consciousness. What we do know is that a range of today's LLMs, symbolic systems, hybrid systems and robots already exhibit features of FC, being able to model themselves and their environments, reason, learn and mimic consciousness-like behaviors, sometimes reporting (or misreporting) that they feel, or are conscious. It is fair to say they may already exhibit some features of FC:

Features of Functional Consciousness (FFC) (df.): *A system has one or more features of functional consciousness when it has some computational features that are at least roughly like those found to resemble conscious processing in humans, (e.g., working memory and attention), or has a capacity to generate behavioral outputs mimicking human linguistic behaviors about consciousness and emotion in ways that can be convincing to some, or an avoidance of situations that a conscious being may seek to avoid because they are painful.*[6]

We've previously discussed FC. FC, in contrast to FCC, involves a more complete set of features, (usually a group of features that are identified by a particular information processing theory of consciousness, e.g., the global workspace theory, to underlie PC). We can think of FC as a more complete and comprehensive set of features that aims to be a candidate for the information processing basis of consciousness, in both biological systems and, possibly, machines, with the proviso that FC in machines may ultimately take others forms related to architectures that differ substantially from those found in biological systems.

An AI that exhibits FFC can simply exhibit one or a few features of consciousness that even laypeople would recognize, without having a particular theory of consciousness in mind. Examples are intelligence, goal directedness and even an understanding of the self/other distinction — features that we associate with sophisticated cases of consciousness in the biological case, such as human consciousness and consciousness of primates. Since we are used to a correlation between FFC and PC in the case of biological entities, it is easy to assume that when we see one or more such features, there is PC in artificial systems. But remember the Zombie LLM thought experiment. If today's LLMs are zombies, as the present authors suspect, then LLMs are cases exhibiting FFC without PC (Schneider, 2019). In this case, FCC-like behavior(s) may be an artifact of a model's training history because the model has been trained on so much human data, including our data about selves, minds and brains, that it behaves like us (see Schneider, 2025).

The deep problem is that just as we have no complete and uncontroversial understanding of FC, we do not currently know what FFC properties, if any, are, necessary or sufficient for phenomenal consciousness. A substance dualist might contend, for example, that an artificial physical system with FCC properties cannot support consciousness unless it is somehow bound to the non-physical element upon which consciousness is dependent. Could we at least regard having one or more FFC properties as being an indication that the

[6] FC is not the same as what philosophers call "functional duplication" or "functional isomorphism." A functional isomorph or duplicate of a system S is a system with all the same functional features as S.

artificial system has PC, much like a biomarker might be used in clinical medicine to indicate the presence of a specific condition (e.g., elevated cardiac troponin as an indicator of damage to the heart)? In principle, yes. However, a widely noted problem with LLMs is that they have been trained on human data, so their reports about consciousness could be a simple feature of their manifesting a FCC by regurgitating or parroting human perspectives on consciousness, just as they achieve other human cognitive capacities like word in context and grounded conceptual mappings when they are scaled up (Wei et al., 2022). While such systems should still be examined for consciousness, this behavior does not constitute evidence for or against consciousness (Schneider, 2025).

Takeaway:

- A system has a feature(s) of functional consciousness (FCC) when it has some computational features that are at least roughly like those found to resemble conscious processing in humans,including reporting consciousness and emotion It is not a straightforward matter to infer that systems having one or more observable FCC features have definitive markers for PC.

Assuming Computational Functionalism's Relative Popularity Extends to Functionalism about Consciousness, in Particular

Some favor the theory that FC is sufficient for C. Debates on this issue concern a larger constellation of issues involving *functionalism*. Functionalism is a view which, in broad strokes, says that a system has a mind when it has a suitable functional organization — a certain causal organization between system inputs, internal states, and system outputs. Mental states are characterised by the causal role they play in the system: how they cause, and are impacted by, their relation to system inputs, outputs and other mental states. When applied to conscious mental states, in particular, functionalism says conscious mental states are characterized in this way (Putnam, 1967; Kim, 2005).

Functionalism, by itself, need not involve computation. However, one influential version of functionalism is called *computational functionalism*. This view says that computational organization, in particular, is what matters for mindedness, and relatedly, mental states' are determined by the computational role they play in the larger computational system, where the kind of computation involved is digital computation (Putnam, 1967; Schneider, 2011).[7]

A word of warning, however: Whereas many cognitive scientists and philosophers of mind support some version of computational functionalism, *many philosophers, even functionalists, did (or do) not support functionalism as a basis for consciousness.* For example, Jerry Fodor was a key figure who framed and defended functionalism, but he remained noncommittal about consciousness. Another key functionalist, Jaegwon Kim, ultimately rejected functionalism as an explanation for consciousness, in particular (Kim, 2005). And Ned Block, a formative functionalist, offered several influential objections to the role of functionalism in supporting consciousness, in particular, identifying a notion of 'access consciousness', akin to FC, and saying that it was conceptually and actually distinct from phenomenal consciousness (Block, 1978).

So if an advocate of AI consciousness claims their view is supported by a larger consensus about functionalism, it is important to realize that there has historically been far less support for computational functionalism about consciousness, even by the leading computational functionalists.

Takeaways:

- "Functionalism as a basis for consciousness" maintains *both* (1) and (2):
  (1) Weak Computational Functionalism: functional mimicry of the processes that support consciousness in human brains can be, in theory, directly and programmatically, implemented in machines in which appropriate concatenations

[7] Digital computation may be insufficient, however. The brain itself uses both digital and analog computation, for example.

of state changes causally link inputs, computationally instantiated mental or cognitive phenomena, and outputs (with or without a "body").

(2) Phenomenal consciousness can be achieved in this way.

- While certain AI systems can be produced that mimic functional capabilities underlying consciousness (2) is very controversial.

Sentience versus Consciousness

As noted, the expression "phenomenal consciousness" can be used synonymously with "sentience," but others can use the expression to refer to those capable of having valenced states, in particular (Birch, 2024). Conversations on AI sentience can risk ambiguity without distinguishing the sense of "sentience" involved. It is an open question whether an entity can be conscious without having valenced states. Philosophers have called such hypothetical cases of conscious beings without valenced states 'Vulcans.' There are vivid examples in nature akin to this phenomenon, although they may not be precise examples. A famous case is that of Jill Bolte Taylor, a physician who had a severe stroke and subsequently reported that during and shortly after the incident she had experience but without a sense of selfhood, and without feeling that something bad or injurious was happening to her (Taylor, 2009). Patients on morphine may feel pain but without valence; perhaps a practiced meditator can achieve consciousness without affect. This controversy is beyond the scope of this piece, but it may suggest that valenced states require a self representation that is actively integrated with perceptual experiences.

Life Versus Consciousness

Members of the public have increasingly become concerned that chatbots, such as GPT 4.0, are "alive." This is likely because their humanlike conversational abilities, including their occasional expressions of emotion and consciousness, remind them of living beings. This raises the larger issue of the relationship between consciousness and life, a complex issue because both concepts are plagued by controversies over formal definitions, whether they can be realized in different substrates, and how to discover "alien" forms. The matter of how to define life is a topic of heated debate in the field of astrobiology, in particular. For the purpose of clarity, we adopt the convention that many astrobiologists do, and utilize the definition adopted by the NASA Astrobiology Institute which is the following: "Life is a self-sustaining chemical system capable of Darwinian evolution" (NASA, 2025). This definition is deliberately broad because astrobiologists are concerned with missing possible cases of life that differ from the one known case on Earth. Even assuming this broad definition, AI doesn't easily qualify, however. For while the field of artificial life (A-life) can simulate artificial agents in Darwinian conditions and self-sustaining ecosystems, or with simulated metabolisms, nonbiological AI systems are not primarily *chemical* systems. At best, such AI agents are mere simulations of living entities, not the entities themselves. So something could be conscious, at least in principle, without being alive, according to this definition. For example, it could be the case that a detailed digital emulation of the brain is conscious while not being alive according to the NASA definition. Being conscious does not entail being alive.

What about vice versa? Panpsychists would say that all forms of life are conscious because everything is conscious, at least in the sense of 'microconsciousness', having at least a small level of experience. However, views that reject panpsychism generally do not say this, (short of vitalism, an older position that has few adherents today). A microbe, or a fig, for instance, while alive, are not generally regarded as being conscious entities. In sum: life and consciousness are what philosophers call "conceptually distinct"—life in principle can exist without consciousness, and vice versa.

However, it may turn out to be the case that in the empirical world, the only AIs that are capable of consciousness are ones that are life forms. Some theorists say that being a form of life is a prerequisite for AI consciousness. Embodied and enactive theories often privilege life and its environmental interactions (Thompson, 2007). Biological naturalists may acknowledge that certain bio-computers have a level of consciousness, at least in the biological component. Further theoretical work would be necessary to

establish why consciousness requires life, if it does. Even if all currently known examples of conscious systems are alive, it is not obvious why reproduction, chemical metabolism, self-maintenance, or other features definitive of life should be required for consciousness.

In contrast, computational functionalists may argue that life is not necessary for consciousness, or, if life is required, then AI consciousness is a two-step process: first, life must be artificially created and manipulated, then the special requirements of consciousness added—a longer timeline, perhaps.

Takeaway:

- Life, as defined by the most commonly accepted definition (NASA), and phenomenal consciousness are conceptually distinct.

Consciousness Versus Self-consciousness

Recall that to be phenomenally conscious is to have inner experiences or feelings of any sort; it is for there to be "something it's like" to be you at a particular moment. *Self*-consciousness involves more: an awareness of yourself as a self, or a person, or an experiencer. You might have conscious visual experiences of the text you are reading, but, plausibly, you are not self-conscious of the fact that you are reading text except on those unusual occasions where you think explicitly about the fact that you are reading. Some theorists of consciousness (e.g., Ginsburg & Jablonka, 2019, Merker 2007) hold that many insects have conscious experiences, but it's controversial whether an insect can ever be self-conscious.

Some theories of consciousness connect consciousness closely with some type of minimal or peripheral self-consciousness. For example, Higher Order theories (Rosenthal 2005; Brown 2025) hold that consciousness always involves representation of one's own mental states. According to other theories (Brentano 1974/1973; Kriegel 2009), conscious experience always entails some at least peripheral awareness of oneself as undergoing that experience. Even theories of insect consciousness (Merker 2007; Ginsburg & Jablonka 2019) sometimes require that a conscious animal be aware of a "self" in the minimal sense of being able to distinguish its own body from its environment.

However, there are two things to note about such theories. First, not all theories of consciousness require any kind of self-consciousness, self-representation, or awareness of oneself. For example Global Workspace Theory (Dehaene 2014) and Local Recurrence Theory (Lamme 2010) do not appear to require self-consciousness for consciousness, even in a minimal sense. Second, even theories, like Higher Order Theories, that do posit a connection between consciousness and self-representation will tend to distinguish between explicitly thematized self-consciousness, of the sort you might have when your attention is called to some feature of yourself, and an unthematized, implicit, ongoing background awareness of your own conscious states.

Holding that Panpsychism Entails AI Consciousness

Panpsychism is an ancient view that says that consciousness is present in everything—every grain of sand, every photon, and so on. Everything has at least a small level of consciousness. Brains have a more complex form of consciousness; we might call this "macroconsciousness." It is important to note that when we are asking questions about conscious machines, we are asking whether they have what we might call macroconsciousness, as opposed to smaller levels that panpsychists contend are possessed by everyday objects, and even regions of the brain itself which do not seem responsible for consciousness, such as the cerebellum.

What exhibits macro consciousness? We know that humans, nonhuman primates, elephants, dolphins, and orca whales have all passed the "mirror test," and perhaps some degree of self-awareness (and consciousness?) extends to certain birds or insects. Could AIs possess macroconsciousness, at least in principle? Not only is it unclear what types of entities have macroconsciousness (AI, biological, universe,

etc.), but it is difficult to tell, within a single type, where it starts. For example, does a shrimp or a mollusk exhibit macroconsciousness? Maybe there is no fundamental dividing line, just a gradual, incremental path toward increasing levels of consciousness. Although we cannot locate a firm line where macroconsciousness begins, we can at least identify items on either end: items that are clearly not macroconscious, such as a photon, and items that clearly are, such as a nonhuman primate or a human. So GPT could be conscious in a really basic panpsychist sense, but attributing macroconsciousness to it could be a major error.

III. Theories of Consciousness for AI Consciousness

Many believe that questions involving AI consciousness and sentience cannot be fully understood, much less answered, except by considering particular theories of consciousness, alongside undertaking careful empirical study. However, understanding the nuances of competing theories of consciousness is a daunting task, for even before turning to AI consciousness, in particular, the landscape of theories of philosophical consciousness that have been developed to address consciousness is large, and the details of a theory can involve a good deal of technical philosophical terminology. Indeed, a recent review paper and website catalogues over 300 kinds of theories of consciousness, organized into a "landscape" of ten categories. Materialism is the category with the most theories; it has 12 subcategories: philosophical, eliminative/illusionism, neurobiological, electromagnetic field, computational & functionalism, homeostatic & affective, embodied & enactive, relational, first-order, higher-order, language, and phylogenetic/evolutionary (Kuhn, 2024, 2025).

In what follows we provide a simplified summary of some of the main theories of consciousness. While we cannot exhaustively list the various theories, we hope to cover those weighing in heavily on the debate over AI consciousness/sentience. We also say, for each view, what the holders of the position may maintain about AI consciousness, drawing from what leading advocates have said, or, in some cases, if scholars advocating this view have said little, providing an approximation of what such a view would say. One noteworthy feature is that many views are surprisingly neutral about AI consciousness, with some important exceptions. While biological naturalists tend to reject the idea that consciousness in a nonneural substrate can be built, and computational functionalists tend to be far more optimistic, beyond this, many of the views discussed may simply punt, stating that it depends upon what our best science is capable of. For example, even a theistic substance dualism need not deny that a science could build a substrate that instantiates conscious systems, it is just that its exponents might claim that an immaterial soul correlates with the physical substrate (Foerst, 2004).

We start with a superordinate category:

Monism. All reality consists in exactly one kind of fundamental ingredient. Some views say the ingredient is mental, others physical.

In contrast to monism, dualists claim that reality consists of two fundamental ingredients. After discussing monistic theories we turn to these.

Theories Standardly Conceived of as Versions of Monism:

Physicalist Monistic Theories:

Physicalism. Consciousness is entirely physical, solely the product of biological systems (like brains) and perhaps nonbiological entities. On this view, mental states fully depend on physical states. An influential version of this view draws from neuroscience, claiming that consciousness is a phenomena of brains (and nervous systems). This view is sometimes called 'materialism,' although some physicalists may reject the name because they may regard all forms of matter as mere ripples in the field.

AI Consciousness: Some versions reject it ("biological naturalists," see below). Many physicalists have not weighed in on the issue and may hold that while *it is in principle conceivable that AI could be conscious, it is currently unclear if it is compatible with the laws of nature that conscious AI can be built, or even if it is, it is unclear whether humans will discover the techniques to do so.* Some are more optimistic, such as transhumanist physicalists. Some insist that embodiment is required, or that today's systems and technology are nowhere near what would be required to harbor or support artificial consciousness because they are not sufficiently like the brain, or an embodied mind. Yet, perhaps silicon systems or systems made of other non-neural substrates could, at a future point, be constructed (see pg.X).

Some popular variants of physicalism that are influential in the consciousness debate are covered below:

Computational Functionalism: computational organization, in particular, is what matters for mindedness, and relatedly, mental states are determined by the computational role they play in the larger computational system, where the kind of computation involved is digital computation.

AI consciousness: as discussed, computational functionalists who also advocate functionalism as a basis for consciousness tend to support the possibility of AI consciousness, at least in principle, though they may be skeptical that it could truly be built.

Informational Theories of Consciousness. Increasingly, information itself is viewed as reality's basic ingredient, being more fundamental than matter, energy and spacetime. Relatedly, consciousness is viewed as being informational. One example is the Integrated Information Theory (IIT) (Tononi et al., 2016). Another is the view that we are all in a computer simulation, which is a species of an information processing view and which may ground itself in the view that reality is informational ("the simulation theory").

AI Consciousness. *Artificial consciousness is in principle possible, but it depends upon the view*. Tononi et al. (2016) stressed that feedforward systems, and systems running on GPUs, were not conscious. If we are all in a simulation, then we are, in fact, conscious AIs, but it doesn't follow that everything in a simulation is a conscious AI, because that could require a certain level of complexity, so perhaps LLMs are not, for instance.

Biological Naturalism: consciousness is an intrinsically biological phenomenon and cannot be developed in a non-biological substrate (Searle, 2017; Seth, 2021).

AI Consciousness: Deeply skeptical about AI consciousness, however, perhaps a biological yet artificial form of consciousness could be created.

Eliminativism/Illusionism: traditional elimitivists claimed that mental states, including consciousness, do not really exist, largely because our categorization of such states fails to match up with the properties of natural phenomena (Churchland, 1986). For instance, in 'Quining Qualia', Daniel Dennett urged that there was nothing that meets the definition of 'qualia', where he defined this in a precise manner (which includes PC, but is stronger). One version of eliminativism is called 'illusionism' to underscore the claim that consciousness is an illusion (Frankish, 2016).

AI Consciousness: since consciousness does not exist, AI consciousness is not possible.

Weak Physical Emergentism: One way philosophers approach consciousness is through the lens of emergence (Chalmers, 2008). On a weak physical emergentist view, consciousness is a higher-level property that supervenes on underlying physical processes. Just as complex weather patterns emerge from molecular interactions, or flocking behavior emerges from individual birds, consciousness may emerge from the vast, dynamic interactions of neurons. This view has direct implications for AI. If consciousness is weakly emergent, then in principle it could arise in artificial

systems, provided they achieve the right kind of physical and functional organization. On this account, there is no mysterious ingredient beyond physical complexity itself - though the scale and type of complexity required remain unknown. The weak emergentist perspective thus offers a naturalistic path for thinking about artificial consciousness, without requiring dualism or nonphysical properties.

> AI Consciousness: If weak physical emergentism is correct, AI consciousness is possible in principle, but only if artificial systems replicate the right kinds of organizational complexity that underlie conscious experience in humans and other animals.

Mysterianism: Mysterians say that the human brain is unable to understand the nature of consciousness and other complex longstanding philosophical problems, largely due to our evolutionary origins and limited cognitive abilities, which were shaped by our history as hunter-gatherers (McGinn,1989).

> AI Consciousness: it would seem difficult to see how this position would assert humans could create conscious entities if they cannot understand it. Perhaps they can reverse engineer it by making biologically based AI, but remain perplexed as to the underlying biological mechanism of consciousness.

We are leaving aside various subtleties. For example, one could be a mysterian and simply be agnostic on physicalism, insisting that it is beyond human understanding to even claim physicalism is correct. (Indeed, philosophers are quite ingenious in locating new views within conceptual space.) We now turn to views that reject physicalism yet are still monistic:

Non-physicalist Monistic Theories:

Idealism. Mental phenomena are the one sole ingredient to reality; everything that seems "physical" is ultimately made up of nonphysical ingredients (Pearce and Goldschmidt, 2018).

> AI Consciousness. *In principle, yes.* If reality just consists in conscious experience, then perhaps AIs can be built that instantiate the level of complexity of macroconscious systems.

Panpsychism. Fundamental physics has phenomenal properties and mentality is a basic ingredient of the universe.[8]

> AI Consciousness. *in principle, yes, but unclear if it could be built*. Consciousness is pervasive, woven into the fabric of reality—but the well-known "combination problem" looms, asking, in this context, how can technology harness micro-experiences to produce macro-minds with unified conscious experiences?

Quantum Theories of Consciousness. Quantum phenomena plays a necessary (perhaps sufficient) role in consciousness, beyond its generic role in all matter. Perhaps it involves microtubules (Hameroff, 2022) or the patterns of decoherence of quantum states in the brain toward 'basins of attraction', impacting mental states directly (Schneider and Bailey, 2025) Note: these views can be physicalist or non-physicalist. Both Hameroff and superpsychism appeal to

[8] Three caveats here: (1) An important exception is Galen Strawson's version of panpsychism, which he believes is physicalist, because Strawson points out that the fundamental building blocks of physical reality/theories are panpsychist, this is part of physical theory (Strawson, 2017). (2) There can be dualist views that take consciousness as a nonemergent property of the universe; on these views only some fundamental particles, not all, are experiential. (3) Panprotopsychism is a related view, which holds that reality is Panprotopsychist, having quasiexperiential ingredients (Chalmers, 2013).

panpsychism, which is most commonly an anti-physicalist view, so we put quantum theories in this superordinate category.

> AI Consciousness. *In principle, yes.* If quantum features are constitutive of consciousness, quantum computers could make quantum theories the lead category for AI consciousness, or perhaps even certain of today's (non-quantum) computers have quantum properties that enable consciousness.

Dualisms:

In contrast to monism, dualists claim that reality consists in two fundamental ingredients. It has two main subtypes:

Substance Dualism. Reality consists in two radically distinct substances—the physical realm, and a separate nonphysical substance (Descartes, 1641). "Emergent dualisms" have agnostic formulations in which physical phenemona ground emergent substances, and psychophysical laws automatically generate consciousness from certain kinds of complexity (e.g., Lowe, 1992).

> AI Consciousness. *It depends.* A radically separate, nonphysical substance may seem to bar AI consciousness, unless the AI itself is correlated with a non-physical substance, the way that the body/brain is.

Property Dualism: all substances are physical, however, reality contains both physical and nonphysical features. Consciousness is a fundamental nonphysical feature of reality, alongside physical properties. A popular variant of this view is emergent property dualism, which says that consciousness is an emergent higher level property instantiated by physical substances (Chalmers, 1998).

> AI consciousness: It depends. This view is amenable to AI consciousness insofar as nonphysical consciousness properties can be instantiated by artificial systems.

Takeaways:

- Questions of sentience, including (especially) AI consciousness, cannot be fully understood, much less completely answered, except in the light of particular theories of consciousness, together with careful empirical investigation.
- It would be an oversimplification at best, and probably a mistake, to assume that materialistic/physicalistic theories are the only legitimate theories of consciousness. Nonphysicalist theories, such as panpsychism, can be compatible with, and draw from, scientific evidence about consciousness.
- It is instructive to assess the prospects of AI consciousness under diverse theories of consciousness (both in principle and in practice).One noteworthy feature is that many views are in principle surprisingly neutral about AI consciousness.
- While biological naturalists tend to reject the idea that consciousness in a nonneural substrate can be built, and computational functionalists about AI consciousness tend to be far more optimistic, beyond this, holders of many of the views discussed can simply say: it depends upon what our best science is capable of.

IV. Summary and Conclusion

The matter of AI consciousness has become pressing. LLMs of increasing linguistic sophistication have at times claimed emotions and consciousness. There is also increasing scientific interest in the field of

machine consciousness as a means of producing more intelligent AI systems, for AI safety research, for BMI development, and simply to produce features of functional consciousness that enable more convincing chatbot interactions. Yet misunderstandings involving machine consciousness are commonplace. Even the core concept of consciousness is easily misunderstood, being commonly conflated with functional consciousness alone which lacks "sensed" or "phenomenological" experience. Such misunderstandings could lead academics, policymakers and the public to miss cases of genuine phenomenal consciousness in AIs ("mindcrimes"), or, conversely, lead to "overattribution" of consciousness to machines or instantiated computer programs. Overattribution can cause unneeded public outcry about "robot rights" and may lead humans to incorrectly view AIs as sentient beings and bona fide relationship companions.

The present paper has sought to clarify the conceptual underpinnings of the debate over AI consciousness (Part One) as well as delve into various sources of conceptual misunderstanding (Part Two). A summary of our takeaways is produced below. Moving forward, it is important to seek further consensus about the potential for AI consciousness through public forums and academic research. Because there is no single commonly agreed upon theory of consciousness, this field must proceed to develop a toolkit of consciousness tests without assuming a particular theory is correct or completely captures all of the nuances of the mechanistic underpinnings of consciousness. Scientific consensus on the components of a panel of tests designed to ascertain whether an artificial entity is truly conscious in the phenomenal sense is an ambitious goal that should be pursued so that we are prepared to administer such a "litmus test" in equivocal cases. It is likely that there will be no "one size fits all" test, given the different kinds of AI systems and controversies at the theoretical level. Alongside the careful development of testing, the field must also continue to seek clarity about key concepts in the field and pursue, in a multi-disciplinary manner, a richer theoretical understanding of consciousness using a common lexicon.

Part One: Conceptual Underpinnings:

- The **core concept of consciousness** is inner experience, or phenomenal consciousness (PC), and it has ethical significance because beings that have PC have subjective experience and, insofar as they are capable of sentience, can suffer and feel a range of emotion.
- The **Problem of AI Consciousness** asks if any AIs have PC.
- The **Human Benchmark Problem** says that humans may not be a completely informative benchmark for AI consciousness, indeed, the human case could be misleading.
- For all we know, the answer to the Hard Problem of AI consciousness, could be 'yes', but only in a single type of AI system. Overall, a careful case by case analysis of different AI systems is required, one that is both sensitive to differences with biological systems and seeks to locate exceptions to the biological case, if such exist.

Particular Distinctions:

An AI could be highly intelligent, being capable of perception, learning, abstraction, reasoning, and calculation, but it may be an AI zombie. Indeed, neuroscientists generally agree intelligence can operate without consciousness in phenomena like blindsight, where patients can navigate obstacles they report being unable to see, or in cases of implicit learning, where people acquire complex patterns without conscious awareness of what they've learned.

Intelligence Versus Consciousness

- An AI could be highly intelligent, being capable of perception, learning, abstraction, reasoning, and calculation, but it may be an AI zombie.
- Neuroscientists generally agree intelligence can operate without consciousness.

Sentience Versus Sapience

- A highly intelligent AI (a system with "sapience") might lack any sentience whatsoever, or have a very minimal degree, say, because it only has a subsystem that has a slight degree of sentience.

- Conversely, a very simple artificial system might have basic sensory experiences, perhaps akin to that had by something like a worm, without sapience.
- Sentience grounds concern about a being's suffering and welfare; sapience alone might ground different kinds of intrinsic moral consideration or none at all. If it does ground intrinsic moral consideration it would be related to properties of sapience such as autonomy or agency.

AI Agency Versus Human Agency

- AI agency is different from philosophical agency.
- Claims of intrinsic moral value of AI agents based on philosophical agency may rest on a conflation and require close scrutiny.

"Functional Consciousness" (FC) Versus "Phenomenal Consciousness" (PC)

- FC and PC are distinct. If they are conflated we risk admitting AI zombies as conscious beings.
- Yet functional consciousness can be expected to track with the level of intelligence of a system. This means FC is of import to AI safety, even if PC is not present.

Seemingly Conscious AIs (SCAICs) Versus Phenomenally Conscious (PC) AIs:

- SCAIC's — "seemingly conscious AIs", are AI's that that seem to have PC but which lack PC.
- Some believe that the issue of AI consciousness is not worth discussion, however, even if today's LLMs are merely SCAICs a sensible scientific and philosophical discussion of the likelihood of consciousness/sentience can help clarify the issues, and prepare for a time in which AI consciousness is reached (if it ever is), and avoid missing or over-attributing sentience.

Features of Functional Consciousness (FCC) Versus Phenomenal Consciousness

- A system has a feature(s) of functional consciousness when it has some computational features that are at least roughly like those found to resemble conscious processing in humans, (e.g., working memory and attention), or has a capacity to generate behavioral outputs mimicking human linguistic behaviors about consciousness and emotion in ways that can be convincing to some, or an avoidance of situations that a conscious being may seek to avoid because they are painful.
- It is not a straightforward matter to consider systems having FCC as having a marker for PC. (E.g., LLMs have been trained on human data, so their reports about consciousness and even similarity in processing activity involving consciousness could be a simple feature of this.)

Assuming Computational Functionalism's Relative Popularity Extends to Functionalism about Consciousness, in Particular

- "Functionalism as a basis for consciousness" maintains *both* (1) and (2):

  (3) Weak Computational Functionalism: functional mimicry of the processes that support consciousness in human brains can be, in theory, directly and programmatically be implemented in machines in which appropriate concatenations of state changes causally link inputs, computationally instantiated mental or cognitive phenomena, and outputs (with or without a "body").

  (4) Phenomenal consciousness can be achieved in this way.

- While certain AI systems can be produced that mimic functional capabilities underlying consciousness (2) is very controversial.

Life Versus Consciousness

- Life, as defined by NASA (2025), is: ”Life is a self-sustaining chemical system capable of Darwinian evolution.”
- In principle, given this definition, an AI system can have PC without being alive, in this sense. And something can be alive (e.g., microbial life) without having PC.

Theories of Consciousness

- Questions of sentience, including (especially) AI consciousness, cannot be fully understood, much less completely answered, except in the light of particular theories of consciousness, together with careful empirical investigation.
- It would be an oversimplification at best, and probably a mistake, to assume that materialistic/physicalistic theories are the only legitimate theories of consciousness, and nonphysicalist theories, such as panspychism, can be compatible with, and draw from, scientific evidence about consciousness.
- It is instructive to assess the prospects of AI consciousness under diverse theories of consciousness (both in principle and in practice).One noteworthy feature is that many views are in principle surprisingly neutral about AI consciousness.
- While biological naturalists tend to reject the idea that consciousness in a nonneural substrate can be built, and computational functionalists about AI consciousness tend to be far more optimistic, beyond this, many of the views discussed can simply say: it depends upon what our best science is capable of.

Glossary

(See the *Theories of Consciousness* section for more terms).

Philosophical “zombies” are entities having the same physical properties of you, yet which lack phenomenal consciousness;

AI zombies: AIs that emulate functional features of consciousness, such as working memory, reportability and attention, but have no felt experience.

The hard problem of consciousness: a problem that asks, “why does all this information processing need to feel a certain way, from inside?” (Chalmers, 1996).

The Problem of AI Consciousness*:* AI may eventually solve problems that even the brightest humans are unable to solve, but still, being made of a different substrate, and being so unlike biological beings, would an AI experience the world?

Phenomenal consciousness (PC): The inner, felt quality, of experience.

The core concept of consciousness: phenomenal consciousness or inner experience, as opposed to functional consciousness.

Sentience: the capacity for subjective experiences, and sometimes it is used synonymously with consciousness. But it is also often used to refer, in particular, to systems that have valenced experiences like pleasure and pain. A sentient being can suffer and feel a range of emotions, such as joy and longing for a better future, and these valenced states matter deeply to that being, so sentience is key to a being’s having intrinsic moral value—moral value that is not merely derivative on the needs or moral value of something else.

Sapience: intelligence and other higher cognitive abilities like the capacity for abstract reasoning, goal planning, and problem-solving.

Global Workspace Theory: consciousness is correlated with an integrative, serial, and slow central workspace in which one's attention and working memory are focused for the purpose of deliberative reasoning, broad ranging searches, and all-things-considered judgements.

Functional consciousness (FC) is an expression that has been used to refer to the information-processing aspect of consciousness.It involves the function of consciousness (i.e., what it does), such as its observable behaviors and causal role within the larger cognitive system (e.g., working memory, attention, decision-making). Modeling FC has been the main interest of many AI consciousness projects as well as the focus of information processing theories in cognitive science, such as the global workspace theory.

Features of Functional Consciousness (FFC): A system has features of functional consciousness when it has some computational features that are at least roughly like those found to resemble conscious processing in humans, (e.g., working memory and attention), or has a capacity to generate behavioral outputs mimicking human linguistic behaviors about consciousness and emotion in ways that can be convincing to some, or an avoidance of situations that a conscious being may seek to avoid because they are painful. FFC is not the same as what philosophers call "functional duplication" or "functional isomorphism," which require an exact parallelism of computational features that are determinative of consciousness.

Seemingly Conscious AIs (SCAICs): AIs that behave as if they are conscious but which are not, because they lack phenomenal consciousness (PC).

Functionalism: a view which says that a system has a mind when it has a suitable functional organization — a certain causal organization between system inputs, internal states, and system outputs. Mental states are characterised by the causal role they play in the system: how they cause, and are impacted by, their relation to system inputs, outputs and other mental states. When applied to conscious mental states, in particular, functionalism says conscious mental states are characterized in this way.

Computational Functionalism. Functionalism, by itself, need not involve computation, but one influential version of functionalism is *computational functionalism*. This view says that computational organization, in particular, is what matters for mindedness, and relatedly, mental states' are determined by the computational role they play in the larger computational system, where the kind of computation involved is digital computation

Functionalism about consciousness: a view that maintains *both* (1) and (2):

> (3) Weak Computational Functionalism: functional mimicry of the processes that support consciousness in human brains can be, in theory, directly and programmatically be implemented in machines in which appropriate concatenations of state changes causally link inputs, computationally instantiated mental or cognitive phenomena, and outputs (with or without a "body").
>
> (4) Phenomenal consciousness can be achieved in this way.

AI agent: an AI Agent is a system with agential capabilities— one that is capable of engaging in independent actions, such as attacking networks, deciding autonomously to engage in defensive or offensive measures, operating without continuous oversight of humans and dynamically adapting to threats, being able to circumvent or override human efforts, potentially acting on its own in high-stakes environments.

Panpsychism: consciousness is present in everything—every grain of sand, every photon, and so on. Everything has at least a small level of consciousness. Brains have a more complex form of consciousness; we might call this "macroconsciousness."

Life: "Life" is a controversial term in the field of astrobiology, but many use The NASA definition of life as the following: "Life is a self-sustaining chemical system capable of Darwinian evolution"

Microconsciousness: according to panpsyhism, everything—every grain of sand, every photon, and so on. Everything has at least a small level of consciousness, which we call microconsciousness to distinguish from macroconsciousness.

Macroconsciousness: Beings with brains and nervous systems, and perhaps complex AIs, exhibit a more complex form of consciousness, "macroconsciousness." It is important to note that when we are asking questions about conscious machines, we are often asking whether they have what we might call macroconsciousness, as opposed to smaller levels that panpsychists contend are possessed by everyday objects, and even regions of the brain itself which do not seem responsible for consciousness, such as the cerebellum.

Vulcans: hypothetical entities said to experience qualia (the hues of a sunset, the sound of an oboe), but without these experiences feeling good or bad to them. Philosophers debate whether creatures they call "Vulcans" deserve intrinsic moral value, as beings with valenced states do, however.

Valenced states: states of experience that feel good or bad to the entity having them (e.g., the painfulness of stubbing one's toe, the joy of hearing one's favorite piece of music, the burning drive of curiousity). Sentient beings have valenced states, Vulcans and AI zombies do not.

Author Biographies

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Robert Lawrence Kuhn: Robert Lawrence Kuhn is the creator, writer, host, and executive producer of *Closer To Truth*, the long-running PBS/public television series and leading global resource on Cosmos (cosmology/physics, philosophy of science), Life (philosophy of biology), Mind (consciousness, brain/mind, philosophy of mind), and Meaning (theism/atheism/agnosticism, global philosophy of religion, critical thinking). Peter Getzels is *Closer To Truth* co-creator and producer/director. Kuhn's comprehensive review article on theories of consciousness – "A Landscape of Consciousness: Toward a Taxonomy of Explanations and Implications" – is published in Progress in Biophysics and Molecular Biology (August 2024, pp. 28-169). In its review, IAI News says, "Kuhn has written perhaps the most comprehensive article on the landscape of theories of consciousness in recent memory." Kuhn has written or edited over 30 books, including The Mystery of Existence: Why is there Anything At All? (with John Leslie); Closer To Truth: Challenging Current Belief; Closer To Truth: Science, Meaning and the Future; The Library of Investment Banking; How China's Leaders Think (featuring President Xi Jinping); The Man Who Changed China: The Life and Legacy of Jiang Zemin (China's best-selling book in 2005 and in December 2022); and "The Origin and Significance of Zero: An Interdisciplinary Perspective" (with Peter Gobets). Dr. Kuhn is chairman of The Kuhn Foundation. He has a BA in Human Biology (Johns Hopkins), PhD in Anatomy/Brain Research (UCLA), and SM (MBA) in Management (MIT).

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Susan Schneider: Dr. Susan Schneider is a philosopher who is the founding director of the Center for the Future of AI, Mind and Society and the William F. Dietrich Eminent Professor at Florida Atlantic University. She formerly held the NASA-Baruch Blumberg Chair with NASA and the Distinguished Scholar Chair at the Library of Congress. Her specializations include philosophy of mind, philosophy of AI, metaphysics and philosophy of science (astrobiology, philosophy of physics, cognitive science). She holds a joint appointment in the Stiles-Nicholson Brain Institute and is a member of FAU's Center for Complex Systems. In her book, *Artificial You: AI and the Future of the Mind*, she discusses the philosophical implications of AI, and, in particular, AI consciousness and the enterprise of "mind design." Schneider recently completed a three year project with NASA on the future of intelligence. She is a co-director of the MPCR Lab at FAU's new Gruber Sandbox, a large facility which builds AI systems drawing from neuroscience research and philosophical developments. She is also the editor of the *Blackwell Companion to Consciousness* (with Max Velmans, Wiley-Blackwell, 2008) and the author of *The Language of Thought: a New Philosophical Direction* (2011, MIT Press). She also appears frequently on television shows on stations such as *PBS* and *The History Channel* and writes opinion pieces for venues such as the *New York Times*, *Scientific American, The Wall Street Journal* and *The Financial Times*.

Eric Schwitzgebel: Eric Schwitzgebel is a Distinguished Professor of Philosophy at, U.C. Riverside and author of four books (most recently *The Weirdness of the World* with Princeton University Press), over a hundred research articles in philosophy and psychology (including in *Philosophical Review* and *Cognition*), a well-known blog, and dozens of opinion pieces (including in the *New York Times* and the *Atlantic*). His mMain research themes are: introspection (esp.ecially our poor knowledge of our own conscious experience), moral psychology (esp. ecially the relationship between philosophical moral reflection and real-world moral behavior, including the moral behavior of professional ethicists), belief (especially in-between cases of believing). His recent work has turned toward AI ethics and AI consciousness, including two books under contract, *AI and Consciousness,* for Cambridge University Press, and *Humanlike: A Defense of AI Rights,* for Princeton University Press.

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